

# ANTHONY DENDER

dukandender@gmail.com

Citizenship: United States and Croatia

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Incoming PhD student in pure mathematics at Pennsylvania State University. Research interests in geometric group theory and metric geometry.

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## EDUCATION

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### Pennsylvania State University

2026-present

- First year Ph.D. student in mathematics
- University Graduate Fellowship recipient

### University of Cambridge, Girton College

2025-2026

- MAST in Mathematics (Part III of the Mathematical Tripos)
- Essay Topic: *Poincaré Duality Groups* (supervisor: Monika Kudlinska)
- Courses examined: *Algebraic Topology, Algebraic Geometry, Commutative Algebra, Geometric Group Theory, Elliptic Curves*

### New York University, New York, NY

2021-2025

- Honors Major in Mathematics, Major in French, Minor in Computer Science
  - GPA: 3.89/4.00
  - Presidential Honors Scholar at the College of Arts and Sciences
  - Studied at NYU Paris and Paris Cité University for the Spring 2024 semester
- Magna cum laude*

### The Park School of Baltimore, Baltimore, MD

Graduated 2021

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## RESEARCH EXPERIENCE

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### Research Project in Metric Geometry

Fall 2024 - Spring 2025

Independent study course with Professor Efe Ok on the topic of metric geometry. Mainly following the texts of Bridson & Haefliger and Petrunin, investigated a particular way in which one could extend concepts in metric geometry (e.g. CAT(k) spaces) into settings in which geodesics in a given metric space are sparse or nonexistent.

### Research Project in Topological Order Theory

Summer 2024 - Fall 2024

Received a grant to participate in NYU's summer research program for undergraduates (SURE), with a project entitled "Order-Embeddability of Topological Posets into Hyperspaces using Non-Canonical Embeddings". Work involved extending the results of a paper by Gerald Beer and Efe Ok on the topic of the existence of topological order-embeddings of topological posets into their hyperspaces.

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## TALKS

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### **"Undergeodesics", Part III Student Seminar**

*December 2025*

*University of Cambridge*

Presented as part of the Part III student seminar series on the topic of extending concepts in metric geometry to settings in which geodesics are sparse or nonexistent.

### **"Order-Embeddability of Topological Posets into Hyperspaces using Non-Canonical Embeddings", SURE Research Presentation**

*October 2024*

*New York University*

Presented the results of a Summer 2024 research project investigating the conditions under which there exists a topological-order embedding of a given topological poset into its hyperspace.

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## READING GROUPS

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### **Algebraic Geometry Seminar (Hodge Theory)**

*Fall 2024*

Participated in a small student seminar covering Hodge theory, following the text *Hodge Theory and Complex Algebraic Geometry I* by Claire Voisin.

### **Algebraic Geometry Seminar (Varieties)**

*Fall 2023*

Participated in a small student seminar on classical algebraic geometry, following the text *Basic Algebraic Geometry 1: Varieties in Projective Space* by Igor Shafarevich.

### **Optimal Transport Seminar**

*Spring 2023*

Worked with a small group of mathematics graduate students and advanced undergraduates in an independent reading group/seminar on the topic of Optimal Transport. The goal of the seminar was to read, discuss, and present about Cédric Villani's text "Topics in Optimal Transportation".

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## EMPLOYMENT HISTORY

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### **Grader, Differential Geometry and Topology**

*January-May 2025*

New York University

- Course grader for the Differential Geometry (MATH-UA 377) and Topology (MATH-UA 375) undergraduate courses at New York University.

### **Transformational Computing Intern, Device Theory Team**

Northrop Grumman Corporation

*June-August 2023*

- Used representation and operator theory to increase the capabilities of simulations of quantum circuit elements. Worked in particular with operator representations of various matrix Lie algebras.

### **Transformational Computing Intern**

Northrop Grumman Corporation

*June-August 2022*

- Worked on developing meshing algorithms for 3D modeling of superconducting electronics.
- Implemented a system for streamlining the process of displaying 3D models of chip data.

## Teaching Assistant

Johns Hopkins Engineering Innovation

*June-August 2021*

- Worked with the instructor and assistant instructor to plan classes, help students, and grade assignments.
- Topics taught included physics, programming, statics, basic statistics, and chemistry, among others, with a general focus on applications to engineering.

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## HONORS / AWARDS

- Dean's List, New York University (2021-2025)
- Dean's Undergraduate Research Fund (DURF) grant recipient (2023)
- Summer Undergraduate Research Experience (SURE) grant recipient (2024)

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## ADDITIONAL SKILLS

- Proficiency in LaTeX
- Intermediate knowledge of Python and Java
- Basic knowledge of MATLAB, Lean, and R
- Professional working proficiency in French
- Basic proficiency in Croatian and Finnish